

Numpy assignments

1. Using numpy create a matrix of size 3 by 5. Create another matrix of 5 by 3
Perform following operations on the matrices
 1. Display shape of both matrix
 2. Find matrix multiplication
 3. Create another matrix of size 3 by 5 using random numbers
 4. And display addition subtraction and member wise multiplication
2. accept 5 numbers from user and store it in a list1
accept 5 numbers from user and store it in list2
accept 5 numbers from user and store it in a list3
accept 5 numbers from user and store it in list4
convert these list into ndarray
combine list1 and list 2 in 1D array, arr1
combine list3 and list 4 in 1D array
combine these 4 arrays into 2D arrays (use stack functions)
3. Store yearwise sales of 2 dealers for tv, freeze, mobiles, washing machines, AC and dishwashers for years (2001 to 2010)

DEALER1

	tv	freeze	mobiles	washing machines	AC	dishwashers
(2001) 0	10000	2000	30000	4000	2000	3000
(2002) 1	10000	2000	30000	4000	2000	3000

Answer the following

- a. Find yearwise sales of tv,freeze,mobile,..... product for both dealer.
- b. Find in each year how many products are sold by dealer1 and how many products are sold by dealer 2.
- c. Display average sales of a 2005 year for each dealer.
- d. Display max sales of a 2005 year for each dealer
- e. Display sales of TV and freeze for each dealer.
- f. Draw 2 bar chart one for each dealer, for each year total number of items sold in the year
- g. Draw a pie chart one for each dealer, total number of items sold in the year